ERDŐS PROBLEM 308 (ALL-N VARIANT): UNIT-FRACTION INTEGER SUMS FORM AN INITIAL INTERVAL FOR EVERY $N \geq 1$

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1. The problem

For an integer $N \geq 1$, let $R(N)$ denote the set of positive integers k that can be written as a sum of distinct unit fractions with denominators in $\{1, \dots, N\}$:

$$R(N) = \Big\{ k \in \mathbb{Z}_{>0} : \exists\, A \subseteq \{1, \dots, N\} \text{ with } \sum_{n \in A} \frac{1}{n} = k \Big\}, \qquad m_N = \Big\lfloor \sum_{n=1}^{N} \frac{1}{n} \Big\rfloor = \lfloor H_N \rfloor.$$

Clearly every $k \in R(N)$ satisfies $k \leq m_N$. In [2, p. 37] (see [3, 4]), Erdős and Graham asked whether $R(N)$ is an initial interval $\{1, \dots, m\}$ (so that no gap ever occurs below the largest representable integer), and whether $m \in \{m_N - 1, m_N\}$. Croot [1] determined the threshold of representability up to $(1 + o(1))\frac{(\log\log N)^2}{\log N}$, establishing both properties for all sufficiently large N. The remaining open variant in [4] asks whether $R(N) = \{1, \dots, m\}$ holds unconditionally for *every* $N \geq 1$. We prove that it does.

Theorem. *For every integer $N \geq 1$, there exists $m \in \{m_N - 1, m_N\}$ such that $R(N) = \{1, \dots, m\}$.*

2. Overview of the proof

Step 1: Reduction from all $N \geq 1$ to one threshold $W(k)$ per integer $k \geq 1$. Because $R(N) \subseteq R(N+1)$ and $R(N) \subseteq \{1, \dots, m_N\}$, the only way $R(N)$ could fail to be $\{1, \dots, m_N - 1\}$ or $\{1, \dots, m_N\}$ is if some integer $1 \leq k \leq m_N - 1$ were missing from $R(N)$. If for each $k \geq 1$ we find a single threshold $W(k)$ satisfying

$$H_{W(k)} < k + 1 \qquad \text{and} \qquad k \in R(W(k)),$$

then whenever $1 \leq k \leq m_N - 1$ we have $H_{W(k)} < k + 1 \leq m_N \leq H_N$, forcing $W(k) < N$ and hence $k \in R(W(k)) \subseteq R(N)$. Thus every integer up to $m_N - 1$ belongs to $R(N)$, so $R(N)$ is either $\{1, \dots, m_N - 1\}$ or $\{1, \dots, m_N\}$.

Choosing

$$W(1) = 1, \qquad W(2) = 10, \qquad W(k) = \lfloor e^{k+0.4188} \rfloor \quad (k \geq 3)$$

places $H_{W(k)}$ right below $k + 1$: since $H_n = \log n + \gamma + O(1/n)$ with $\gamma \approx 0.5772$, the exponent shift $0.4188 < 1 - \gamma \approx 0.4228$ guarantees both $H_{W(k)} < k + 0.996 < k + 1$ (for $k \geq 3$; in fact $H_{W(k)} \in (k + 0.9958, k + 0.99992)$ for $k \geq 5$) and a surplus $H_{W(k)} - k > 0.9958$ of nearly one full unit of harmonic sum with which to cancel all unwanted prime powers from the denominator.

Step 2: Cancelling prime factors from the denominator within a 0.9921 loss budget. For $1 \leq k \leq 9$, explicit subsets $A_k \subseteq \{1, \dots, W(k)\}$ summing to k are given directly (for instance $1 = \frac{1}{1}$ and $2 = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{6}$).

For $k \geq 10$ ($n = W(k) \geq 33{,}483$), if we start from the full set $\{1, \dots, n\}$, its reciprocal sum $H_n \approx k + 0.996$ exceeds k by almost 1, but its reduced denominator divides $\operatorname{lcm}(1, \dots, n) \approx e^n$. We must trim a subset of denominators of total reciprocal sum < 0.9921 so that *every* prime power outside $11520 = 2^8 \cdot 3^2 \cdot 5$ disappears from the reduced denominator, and then adjust the remaining integer multiple of $1/11520$ down to k:

(1) **Large primes ($p > n/M_{\text{top}}$) and prime squares (p^2 for $p > M_{\text{bot}}$):** Simply omit all multiples of primes $p > n/M_{\text{top}}$ (costing at most $\mathcal{L}_{\text{tail}} = \sum_{m<M_{\text{top}}} \frac{1}{m} \sum_{n/M_{\text{top}}<p\le n/m} \frac{1}{p} \lesssim \frac{1.0386(\log M_{\text{top}})^2}{2\log n}$) and all multiples of p^2 for $p > M_{\text{bot}}$ (costing at most $\mathcal{L}_{\text{sq}} \lesssim \frac{\log n}{M_{\text{bot}}}$).

(2) **Medium primes ($M_{\text{bot}} < p \le n/M_{\text{top}}$):** Multiples of p in the sieved set contribute too much harmonic weight ($\approx \log \frac{\log n}{\log M_{\text{bot}}}$) to discard wholesale. Instead, we *peel* each such prime p modularly using a *modular gadget* G_p: a small set $S_G \subseteq \{1, \dots, M_{\text{bot}}\}$ of divisors of an M_{bot}-smooth number D whose subset sums $\sum_{m\in T} D/m$ hit every integer in a contiguous interval $[A, B]$ of length $\ge p$. Scaling S_G by pk_p ($k_p = \min(G_p.M, \lfloor n/(p\,G_p.M)\rfloor)$), coprimality $\gcd(p, D) = 1$ lets us pick $t_p \in [A, A+p-1]$ in the exact residue class modulo p that makes the p-adic valuation of the remaining rational sum ≥ 0, at reciprocal cost only $\approx \eta/(pk_p)$ ($\eta = \sum_{m\in S_G} 1/m$).

(3) **Higher powers of small primes (q^a for $q \le M_{\text{bot}}$, $q^a \nmid 11520$):** Whenever $q^a \cdot 2^{\lfloor \log_2 q\rfloor} \le n$, instead of discarding all multiples of q^a (which would cost $H_{\lfloor n/q^a\rfloor}/q^a$), we group multiples of q^a by their odd part and use binary subset sums over $\{q^a, 2q^a, 4q^a, \dots\}$ at the single top dyadic scale to reduce v_q one step at a time, cutting the cost down to $O(1/q^a)$ (independent of $\log n$).

(4) **Final 11520-smooth adjustment:** Once all primes > 5 and higher powers $2^9, 3^3, 5^2$ are cancelled, the remaining subset $S_1 \subseteq \{1, \dots, n\}$ has $\sum_{d\in S_1} 1/d = u/11520 \in (k, k+1)$. All 54 divisors of 11520 still lie in S_1, and their scaled reciprocals $\{11520/d : d \mid 11520\}$ represent every integer in $[0, 42466]$ as a subset sum. Since $0 < u - 11520k < 11520 < 42466$, deleting the matching divisor subset leaves $\sum_{d\in A} 1/d = k$ exactly.

3. Bridging Finite Computation and Asymptotics

To cover every $k \ge 10$ with no gap, we establish effective Chebyshev bounds

$$0.9612\,x - 3.44\log x - 4.26\sqrt{x} \le \theta(x) \le 1.0386\,x \qquad (x \ge 300)$$

directly from Stirling's formula and zero-mean integer linear combinations of $\log(\lfloor x/a_i\rfloor!)$, together with Abel summation for $\sum_{a<p\le b} 1/p$:

- **Intermediate regime ($10 \le k \le 149$):** Partitioned into 15 intervals covering $[10, 149]$ ($[10,10], \dots, [18,18]$, $[19,21]$, $[22,25]$, $[26,34]$, $[35,55]$, $[56,99]$, and $[100,149]$). Each interval uses two greedy divisor-extension gadgets built from $D_0 = 2520$, and integer interval arithmetic at scale 10^9 certifies that $\mathcal{L}_{\text{tail}} + \mathcal{L}_{\text{sq}} + \mathcal{L}_{\text{peel}} + \mathcal{L}_{\text{QA}} \le 0.9921$.
- **Asymptotic regime ($k \ge 150$):** Setting $R = \lfloor 4k/3\rfloor + 220$ and building a single binary-lifting gadget on $D = 2^{1+\lfloor \log_2 R\rfloor} \prod_{3\le q\le R} q > 1000e^{k+1} > 1000n$ with smoothness $M = 2R^2 \asymp \frac{32}{9}k^2$ and $\eta \le 1$ yields uniform analytic bounds across all $k \ge 150$:

  $$\mathcal{L}_{\text{tail}} \le 0.790, \qquad \mathcal{L}_{\text{sq}} \le 0.003, \qquad \mathcal{L}_{\text{peel}} \le 0.115, \qquad \mathcal{L}_{\text{QA}} \le 0.024,$$

  whose sum 0.932 lies well inside the 0.9921 budget.

Combining the regimes $1 \le k \le 9$, $10 \le k \le 149$, and $k \ge 150$ proves $k \in R(W(k))$ for every $k \ge 1$, establishing the Theorem for every $N \ge 1$.

Remark. The complete proof is formalized in Lean 4 without custom axioms in the repository file `FormalConjectures/ErdosProblems/308.lean` (theorem `erdos_308.variants.all_N`).

References

[1] E. S. Croot, III, *On some questions of Erdős and Graham about Egyptian fractions*, Mathematika **46** (1999), 359–372.

[2] P. Erdős and R. L. Graham, *Old and new problems and results in combinatorial number theory*, Monographies de L'Enseignement Mathématique, vol. 28, Université de Genève, 1980.

[3] *Erdős Problem #308*, `https://www.erdosproblems.com/308`.

[4] The Formal Conjectures Authors, *FormalConjectures/ErdosProblems/308.lean*, `https://github.com/google-deepmind/formal-conjectures`.